

State Variables of Real Gases

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The isotherms of sulfur hexafluoride (SF_6) were measured for a set of temperatures in between $30^\circ C$ and $50^\circ C$. From these the critical temperature as well as the specific energy of vaporization are determined as $T_c = (49,4 \pm 2,4)^\circ C$ and $E_{vap} = (135,1 \pm 1,8) \frac{kJ}{kg}$. Under assumption of a Van der Waals Gas, the amount of substance, the Van der Waals Constants and the critical exponent are computed as $\nu = (0,00204 \pm 0,00013) mol$, $a = (0,782 \pm 0,038) Pa (\frac{m^3}{mol})^2$, $b = (8,64 \pm 0,40) 10^{-5} \frac{m^3}{mol}$ respectively. A comparison of theory and experiment is made at the $30,8^\circ C$ isotherm. The model is shown to be quantitatively inadequate in the phase coexistence regime.

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1 Introduction

Modern Thermodynamics is a truly fundamental field of research for both the theoretical and the applied sciences[1]. Especially for the latter it is important to have a model which makes accurate predictions for the behaviour of real gases. A common model for such gases is the Van der Waals Gas, which only takes molar volume and attracting forces into account and yet makes adequate qualitative predictions for phase transitions and critical behavior. This model was tested on sulfur hexafluoride (SF_6) since it is both color- and odorless, non poisonous and chemically inert, similar to nitrogen, yet features a relatively easy to examine critical point.

2 Critical parameters of sulfur hexafluoride

Measuring isotherms is a simple method for examining critical behaviour of gases. The experimental setup includes a pressure chamber with adjustable volume in a heat bath (thermostat) (Figure 1). Multiple measurements of the isotherms for different temperatures between $30^\circ C$ and $50^\circ C$ were made, by setting a constant temperature on the thermostat and then decreasing the volume with the pressure regulator. Special attention was paid to the regimes where vaporization and condensation take place. (see Appendix Figure A1)

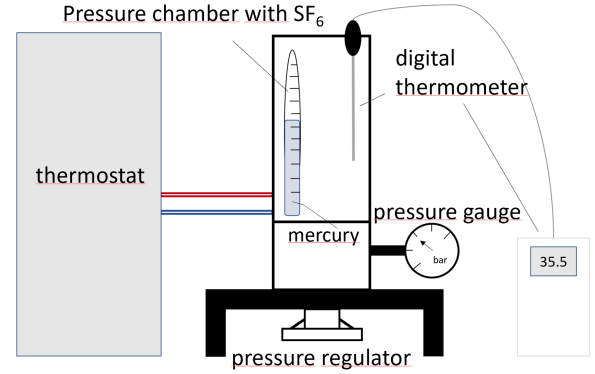


Figure 1: Experimental setup to measure the isotherm pressure for different volume, adjustable by the pressure regulator [2].

2.1 Vapor pressure diagram of sulfur hexafluoride

Under assumption of a Van der Waals Gas the equation of state reads:

$$(p + \frac{\nu^2 a}{V^2})(V - \nu b) = \nu RT. \quad (1)$$

The variables p , V and T are the thermodynamic variables of state known as pressure, volume and temperature. The amount of substance is denoted with ν and the universal gas constant with R . The constants a and b are substance specific and commonly referred to as internal pressure and co-volume respectively. They take into account that the molecules of

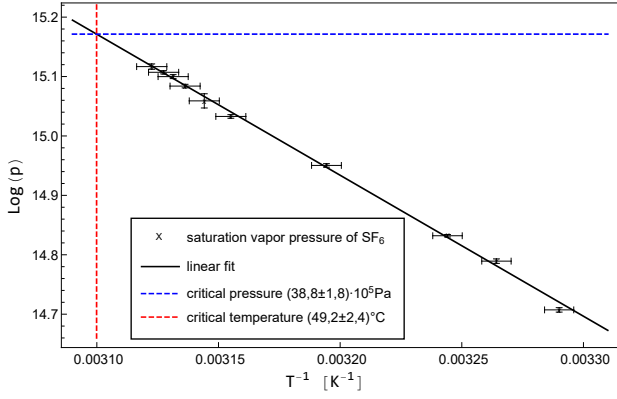


Figure 2: Inverse temperature over the logarithm of the saturation pressure, fitted by a linear polynomial for SF_6 . The horizontal line is the critical pressure, evaluated in Figure A3. The abscissa value given by the intersection of the two lines determines the inverse critical temperature $\frac{1}{T_c}$.

real gases have interactive forces between themselves and a non-zero volume they occupy in space.

In order to obtain the critical pressure p_c and volume V_c the following analysis was employed. The measured isotherms were divided into three linear parts - the part with constant pressure in the middle and the decreasing pressure on the left and right. The isobar is especially important since a linear fit through it is equivalent to the Maxwell Construction, which in a Van der Waals gas describes the pressure of phase coexistence. The construction usually requires finding the isobar which equalizes the area under the $p(V)$ curve as shown in Figure 3, but in this case it's unnecessary since the $p(V)$ dependency is essentially linear.

Each section of the isotherm is now fitted linearly (Figure A2). Each intersection of the horizontal line with the other two is computed. The error of the intersection is determined by greatest deviation of the 1σ confidence bands intersections. The set of all intersections is fitted with a 6th degree polynomial in the interval of the lowest and highest volume and its maximum computed in order to gain the critical point (see Figure A3).

Critical Parameter	Value	Reference value
p_c in [$10^5 Pa$]	$38,8 \pm 1,8$	37,53
T_c in [$^{\circ}C$]	$49,4 \pm 2,4$	45,6
E_{vap} in [$\frac{kJ}{kg}$]	$135,1 \pm 1,8$	153,2

Table 1: Comparison of critical values with their literature values from Ref. [5] and Ref. [6].

This delivers $p_c = (38,8 \pm 1,8) \cdot 10^5 Pa$ and $V_c = (5,29 \pm 0,123) \cdot 10^{-7} m^3$. To gain the critical temperature we plot the logarithm of pressure over the inverse temperature and apply a linear fit (Figure 2). The data shown in Figure 2 are taken from the set of intersections corresponding to starting points of gas condensation at different temperatures. T_c can be obtained by extrapolating the linear fit and taking the value of T at $p = p_c$, thus $T_c = (49,4 \pm 2,4)^{\circ}C$.

We further calculate the vaporization energy. It is the amount of energy needed to transform a given amount of substance from liquid to the gaseous phase. For one molecule the equation reads:

$$E = -mk_b \quad (2)$$

with m being the slope of the linear fit of $\log p(T^{-1})$ (Figure 2) and k_b the Boltzmann constant [4]. This energy divided by the molar mass of SF_6 [5] yields the specific vaporization energy $E_{vap} = (1,351 \pm 0,018) \cdot 10^5 \frac{J}{kg}$.

A comparison of the results with literature values for p_c , T_c and E_{vap} [5], [6] shows, that none of these values agree within their error margins except for the critical pressure. Since the other values are at least of the same order of magnitude as the literature values, one can conclude that no major errors were made during measurement, but rather that the provided SF_6 contained impurities - most likely water.

2.2 Gas-specific parameters

The constants a and b from eq. (1) can be obtained (as shown in [1]) by calculating the theoretical critical values p_c , V_c and T_c and further using equations:

$$a = \frac{27R^2T_c^2}{64p_c} \quad (3)$$

$$b = \frac{RT_c}{8p_c} \quad (4)$$

Inserting (3) and (4) into (1) and solving for ν gives

$$\nu = \frac{8p_cV_c}{3RT_c} \quad (5)$$

Employing the previously determined critical parameters equation (5) delivers $\nu = (0,00204 \pm 0,00013) mol$. One should keep in mind, that this result is gained under the assumption of a Van der Waals gas. Since this model is very simple the calculated amount of substance is not representative for the sulfur hexafluoride in the pressure chamber. This is due to the fact, that the Van der Waals gas is just a poor quantitative approximation, as will be discussed in the next section. Additionally we calculate the numerical parameters a and b using (3) and (4) as $a = (0,782 \pm 0,038) Pa (\frac{m^3}{mol})^2$ and $b = (8,64 \pm 0,40) \cdot 10^{-5} \frac{m^3}{mol}$. Both are in agreement with their literature values [3] - $a_{lit} = 0,7857 \cdot Pa (\frac{m^3}{mol})^2$

and $b_{lit} = 8,79 \cdot 10^{-5} \frac{m^3}{mol}$.

The critical coefficient K_c of real gases is given by:

$$K_c = \frac{RT_c \nu}{p_c V_c}. \quad (6)$$

A comparison with (5) shows, that the critical coefficient is determined by parameters which were calculated under the condition of the critical point. Inserting (5) into (6) gives $K_c = \frac{8}{3}$ - a universal value. We note that any error analysis is redundant, since all faulty values are factored out. Since this calculation of K_c essentially relies on circular reasoning another method would be preferable. One could - for example - determine the amount of substance by weighing.

3 Comparison of theory and experiment

Whether or not the Van der Waals model is valid, or rather under which conditions it is valid can be easily checked by comparing the theoretical and the measured isotherms (see Figure 3). The first conclusion is, that SF_6 does not behave like an ideal gas for low volumes, only for larger ones. This can be explained by the fact, that at larger volumes the intrinsic volume of the particles and their interactions don't matter as much as for low volumes. The Van der Waals model holds true for both low and high volumes, but fails to accurately describe the behaviour in the phase coexistence regime. The real data shows a constant pressure dependence for $p(V)$ in the phase coexistence regime. However the Van der Waals model assumes that the energy gained through compression increases pressure, while in a real gas it is used for the phase transition - thus it predicts a too large critical pressure and temperature.

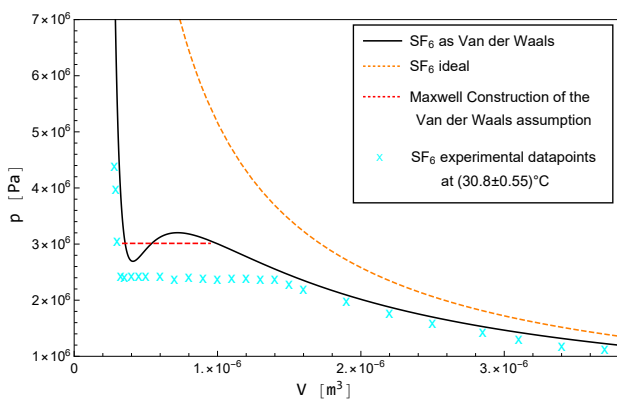


Figure 3: Model comparison for SF_6 . Plotted are the theoretical dependence $p(V)$ for SF_6 as an ideal gas (orange curve), as a Van der Waals Gas with experimentally determined coefficients (black curve), both at $T = 30.8^\circ C$, and the measured data at $(30.80 \pm 0.55)^\circ C$.

4 Conclusion

Twelve isotherms of SF_6 were measured in order to study its critical behaviour. The method of the Maxwell Construction was used to determine the coexistence region of fluid and gaseous phases. Its maximum was identified as the critical point with $p_c = (38,8 \pm 1,8) \cdot 10^5 Pa$ and $V_c = (5,29 \pm 0,123) \cdot 10^{-7} m^3$. The former is in agreement with the literature value of $37,53 \cdot 10^5 Pa$ [5]. Both the critical temperature and the specific evaporation energy were obtained by plotting the logarithm of pressure over the inverse temperature. This led to the values $T_c = (49,4 \pm 2,4)^\circ C$ and $E_{vap} = (1,351 \pm 0,018) \cdot 10^5 \frac{J}{kg}$ which both aren't in agreement with their literature values[5], [6]. This result can be explained by impurities in the used sulfur hexafluoride. Additionally the amount of substance, the internal pressure and the co-volume were computed from the Van der Waals equation and the conditions for criticality - their values are $\nu = (0,00204 \pm 0,00013) mol$, $a = (0,782 \pm 0,038) Pa(\frac{m^3}{mol})^2$ and $b = (8,64 \pm 0,40) \cdot 10^{-5} \frac{m^3}{mol}$. While the two Van der Waals constants agree within the error margin with their literature values for SF_6 [3], one should not take the calculated amount of substance as exact value. The latter was calculated using the critical values obtained under assumption of a Van der Waals Gas, which as Figure 3 and Figure A3 show, is precisely where the model fails to describe the observed behaviour. Taking all observations into account, it is obvious that the Van der Waals model, whilst making rather accurate predictions for lower and higher volumes, fails to quantitatively describe the phase coexistence regime of sulfur hexafluoride.

References

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Appendix

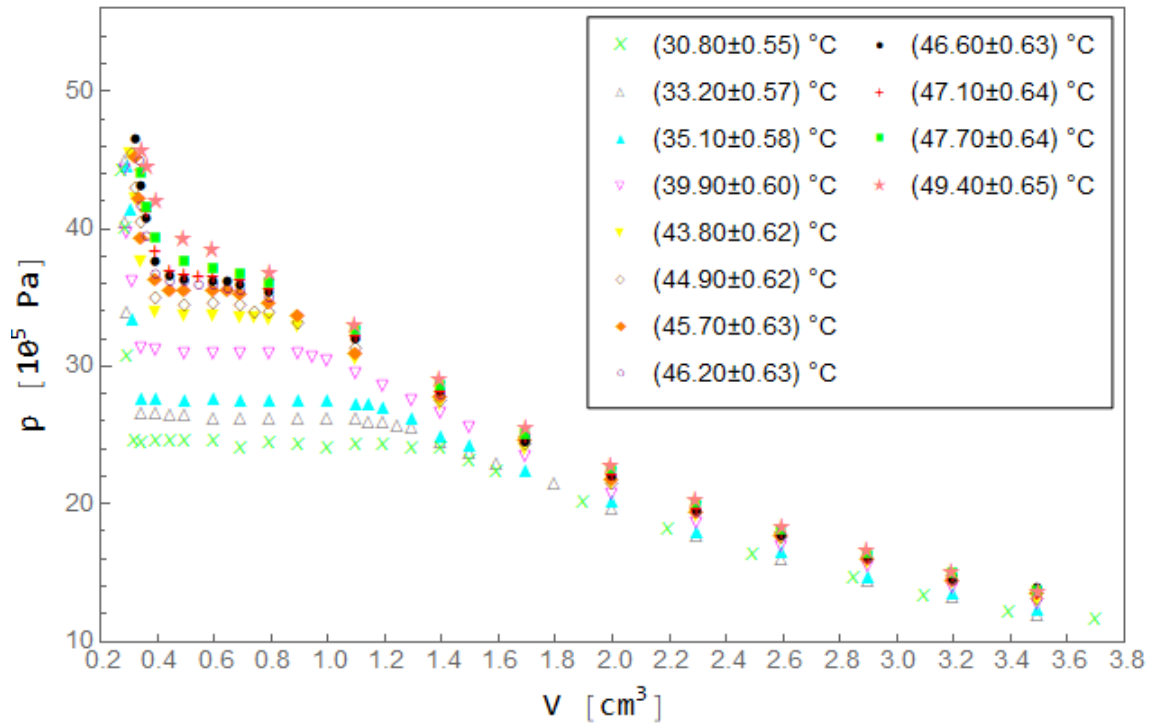


Figure A1: All experimental measured isotherms of SF_6 in a interval of temperature from $(30.80 \pm 0.55)^\circ C$ to $(49.40 \pm 0.65)^\circ C$.

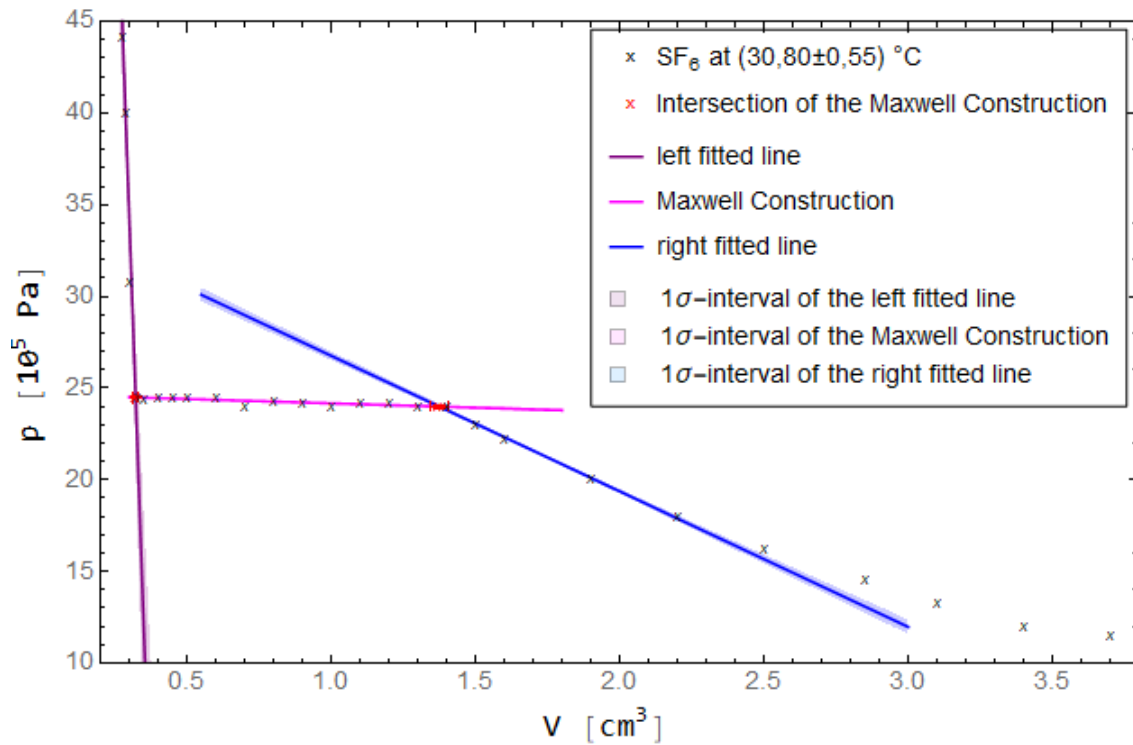


Figure A2: Exemplary method to determine the intersections for the $(30.80 \pm 0.55)^\circ C$ isotherm of SF_6 . The intersections determine the endpoints of the Maxwell Construction. The errors were computed by the intersections of the 1- σ -interval of each linear fit.

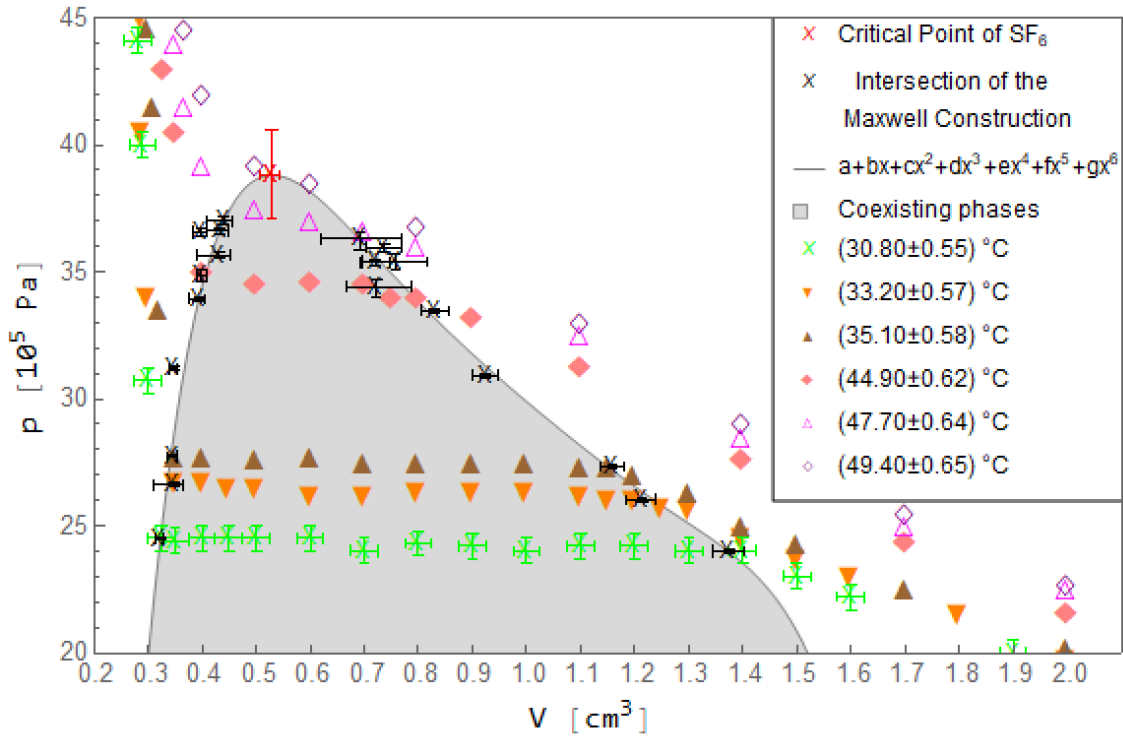


Figure A3: Several isotherms of SF_6 and all intersections of the Maxwell Constructions are plotted together. The area under the 6th order polynomial fit for the intersections shows the coexistence regime. The maximum determines the critical point, i.e. the critical pressure p_c and the critical volume V_c .